

9.2.6 Wireless Network Settings

Wireless Network Enabled - This setting activates and deactivates wireless networking.

Obtain Address Automatically - Lets the machine retrieve an IP address and other network information from the network's Dynamic Host Configuration Protocol (DHCP) server. You can use this option only if your network has a DHCP server.

IP Address - The machine's static TCP/IP address on a network without a DHCP server. Your network administrator assigns this address to your machine.

Subnet Mask - Your network administrator assigns the subnet mask value for machines with a static TCP/IP address.

Default Gateway - An address to gain access to your network through routers. Your network administrator assigns this address.

DNS Server - The name of the Domain Name Server or DHCP server on the network.



NOTE:

The address format for Subnet Mask, Gateway, and DNS is XXX.XXX.XXX.XXX. Do not end the address with a period. Do not use negative numbers. 255.255.255.255 is the highest possible address.

Wireless SSID - The name of the wireless access point. You can enter this manually, or you can press the LEFT or RIGHT cursor arrow keys to select from a list of available networks. If your network does not broadcast its SSID, you must enter this manually.

Wireless Security - The security mode that your wireless access point uses.

Password - The password for the wireless access point.

9.2.7 Net Share Settings

Net Share lets you connect remote computers to the machine control over the network, to transfer files to and from the machine's User Data directory. These are the settings you need to adjust to set up Net Share. Your network administrator can give you the correct values to use. You must enable remote sharing, local sharing, or both to use Net Share.

After you change these settings to the correct values, press **[F4]** to begin Net Share.



NOTE:

If you need special characters such as underscores (_) or carets (^) for these settings, refer to page 60 for instructions.

CNC Network Name - The name of the machine on the network. The default value is **HAASMachine**, but you must change this so that each machine on the network has a unique name.

Domain / Workgroup Name - The name of the domain or workgroup the machine belongs to.

Remote Net Share Enabled - When this is **ON**, the machine shows the contents of the shared network folder in the **Network** tab in the Device Manager.

Remote Server Name - The remote network name or IP address of the computer that has the share folder.

Remote Share Path - The name and location of the shared remote network folder.



NOTE:

Do not use spaces in the shared folder name.

Remote User Name - The name to use to log in to the remote server or domain. User names are case-sensitive and cannot contain spaces.

Remote Password - The password to use to log in to the remote server. Passwords are case-sensitive.

Local Net Share Enabled - When this is **ON**, the machine allows access to the **User Data** directory to computers on the network (password required).

Local User Name - Displays the user name to log into the control from a remote computer. The default value is **haas**; you cannot change this.

Local Password - The password for the user account on the machine.



NOTE:

You need the local user name and password to access the machine from an outside network.

Net Share Example

In this example, you have established a net share connection with the **Local Net Share Enabled** setting turned **ON**. You want to view the contents of the machine's **User Data** folder on a networked PC.

**NOTE:**

This example uses a Windows 7 PC; your configuration may vary. Ask your network administrator for help if you cannot establish a connection.

1. On the PC, click the START menu and select the RUN command. You can also hold the Windows key and press R.
2. At the Run prompt, type (2) backslashes (\\) and then the machine's IP address or CNC Network Name.
3. Click OK or press Enter.
4. Type the machine's **Local User Name** (haas) and **Local Password** in the appropriate fields, and then click OK or press Enter.
5. A window appears on the PC with the machine's **User Data** folder displayed. You can interact with the folder as you would with any other Windows folder.

**NOTE:**

If you use the machine's CNC Network Name instead of the IP address, you may need to type a backslash before the User Name (\haas). If you cannot change the username in the Windows prompt, select the "Use another account" option first.

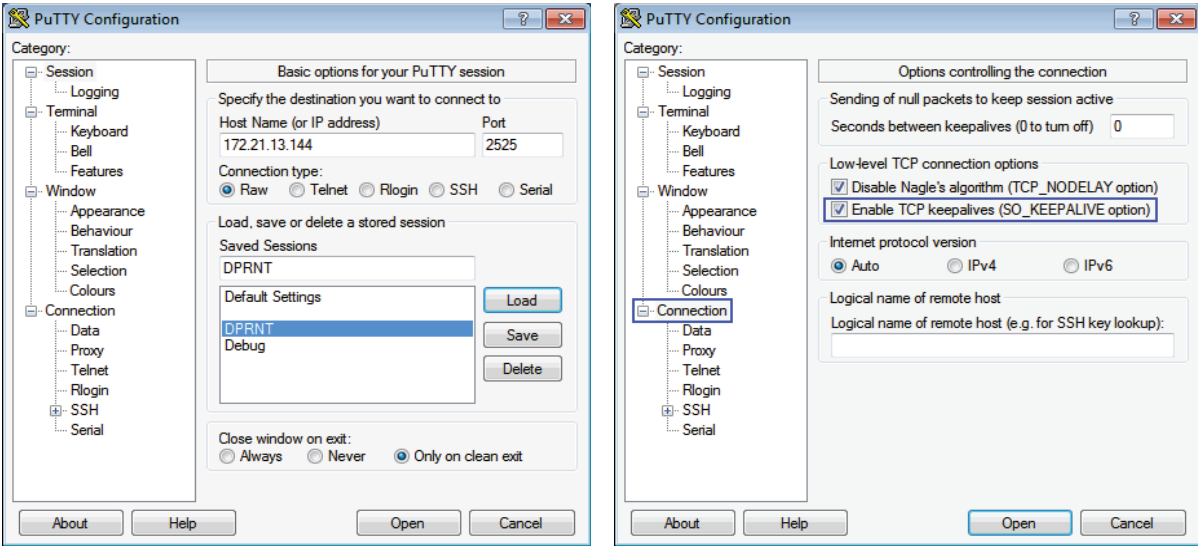
9.2.8 Machine Data Collection

Machine Data Collection (MDC) lets you use Q and E commands to extract data from the control through the Ethernet port or the Wireless Networking option. Setting 143 both enables the feature and specifies the data port that the control uses to communicate. MDC is a software-based feature that requires an additional computer to request, interpret, and store data from the control. The remote computer can also set certain Macro variables.

The Haas control uses a TCP server to communicate over networks. On the remote computer, you can use any terminal program that supports TCP; examples in this manual use PuTTY. Up to (2) simultaneous connections are allowed. Output requested by one connection is sent to all connections.

1. In the basic options section, type the machine's IP address and the port number in Setting 143. Setting 143 must have a nonzero value to use MDC.
2. Select the Raw or Telnet connection type.
3. Click "Open" to start the connection.

F9.19: PuTTY can save these options for subsequent connections. To keep the connection open, select "Enable TCP keepalives" in the "Connection" options.



To check the connection, type `?Q100` in the PuTTY terminal window. If the connection is active, the machine control responds with `SERIAL NUMBER, XXXXXX`, where `XXXXXX` is the machine's actual serial number.

Data Collection Queries and Commands

The control responds to a Q command only when Setting 143 has a nonzero value.

MDC Queries

These commands are available:

T9.2: MDC Queries

Command	Definition	Example
Q100	Machine Serial Number	>Q100 SERIAL NUMBER, 3093228
Q101	Control Software Version	>Q101 SOFTWARE, VER 100.16.000.1041
Q102	Machine Model Number	>Q102 MODEL, VF2D
Q104	Mode (LIST PROG, MDI, etc.)	>Q104 MODE, (MEM)
Q200	Tool Changes (total)	>Q200 TOOL CHANGES, 23

Command	Definition	Example
Q201	Tool Number in use	>Q201 USING TOOL, 1
Q300	Power-on Time (total)	>Q300 P.O. TIME, 00027:50:59
Q301	Motion Time (total)	>Q301 C.S. TIME, 00003:02:57
Q303	Last Cycle Time	>Q303 LAST CYCLE, 000:00:00
Q304	Previous Cycle Time	>Q304 PREV CYCLE, 000:00:00
Q402	M30 Parts Counter #1 (resettable at control)	>Q402 M30 #1, 553
Q403	M30 Parts Counter #2 (resettable at control)	>Q403 M30 #2, 553 STATUS, BUSY (if in cycle)
Q500	Three-in-one (PROGRAM, Oxxxxx, STATUS, PARTS, xxxxx)	>PROGRAM, O00110, IDLE, PARTS, 4523
Q600	Macro or system variable	>Q600 801 MACRO, 801, 333.339996

You can request the contents of any macro or system variable with the **Q600** command; for example, **Q600 xxxx**. This shows the contents of macro variable **xxxx** on the remote computer.

Query Format

The correct query format is **?Q###**, where **###** is the query number, terminated with a new line.

Response Format

Responses from the control begin with **>** and end with **/r/n**. Successful queries return the name of the query, then the requested information, separated by commas. For example, a query of **?Q102** returns *MODEL, XXX*, where *XXX* is the machine model. The comma lets you treat the output as comma-separated variable (CSV) data.

An unrecognized command returns a question mark followed by the unrecognized command; for example, **?Q105** returns **?, ?Q105**.

E Commands (Write to Variable)

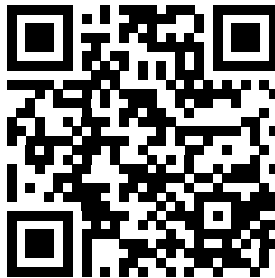
You can use an E command to write to macro variables **#1–33**, **100–199**, **500–699** (note that variables **#550–580** are unavailable if the mill has a probing system), **800–999** and **#2001** through **#2800**. For example, **Exxxx yyyyyy.yyyyyy** where **xxxx** is the macro variable and **yyyyyy.yyyyyy** is the new value.

**NOTE:**

When you write to a global variable, make sure that no other programs on the machine use that variable.

9.2.9 Haas Connect

HaasConnect is a web-based application that lets you monitor your shop with a web browser or mobile device. To use HaasConnect, you set up an account [at myhaascnc.com](https://myhaascnc.com), add users and machines, and designate the alerts you want to receive. For more information about HaasConnect, go to diy.haascnc.com/haasconnector scan the QR code below with your mobile device.



9.2.10 Remote Display View

This procedure tells you how to view the machine display on a computer. The machine must be connected to a network with an Ethernet cable or with a wireless connection.

Refer to the Networking Connection section on page **408** for information on how to connect your machine to a network.

**NOTE:**

You must download the VNC Viewer to your computer. Go to www.realvnc.com to download the free VNC Viewer.

1. Push the **[SETTING]** button.
2. Navigate to the Wired Connection or Wireless Connection tab in the Network tab.
3. Write down the IP Address for your machine.

4. Remote Display Tab

Settings

Settings Network User Positions Alias Codes Connection Wireless Connection Net Share Haas Connect **Remote Display**

Remote Display Status

Remote Display: Up

Remote Display requires a strong password.
A strong password requires 8 characters or more, one upper case letter,
one lower case letter, one numeric digit, one symbol (@#\$%&*).

Name	Value
Remote Display	On
Remote Display Password	*****

Warning: Changes will not be saved if page is left without pressing **F4**

F3 Discard Changes **F4** Apply Changes



NOTE:

The Remote Display tab is available in software version 100.18.000.1020 or higher.

5. Navigate to the Remote Display tab in the Network tab.
6. Turn **on** the Remote Display.
7. Set the Remote Display Password.



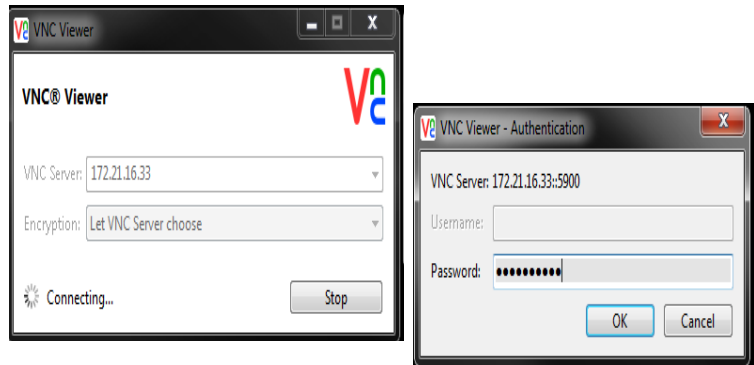
NOTE:

The Remote Display feature requires a strong password, follow the guide lines on the screen.

Press **[F4]** to apply settings.

8. Open the VNC Viewer application on your computer.

9. VNC Software Screen



Enter your IP Address in VNC Server. Select **Connect**.

10. At the login box enter the password you entered at the Haas control.
11. Select **OK**.
12. The machine display shows on your computer screen